

Aoun: Security Verification Report

Two-way LoRa distress and tracking system · Team: D. Alanazi (Leader), R. Aldosari, J. Alsharif, H. Alhejjei, A. Alqarni · Digital Technical College for Girls, Riyadh · September 2026

1. Scope

This report verifies the message-security layer of Aoun (`aoun_crypto.h`) before hardware assembly. The same source file runs on the ESP32 (LILYGO T-Beam) firmware. It was compiled for a PC host against Mbed TLS 3.6.2 and run through the automated test below.

2. Design

- **Cipher:** AES-128-GCM (authenticated encryption), 128-bit key shared by paired devices, 16-byte authentication tag.
- **Nonce:** sender ID + 32-bit sequence number (boot counter kept in flash × message counter), so a nonce is never reused.
- **Header** (version, sender ID, sequence) is authenticated as associated data.
- **Replay protection:** the receiver accepts only sequence numbers higher than the last one seen from that device.
- **Whitelist:** only registered device IDs are accepted. The key is stored outside the public repository.

3. Test results: 10/10 passed

Test case	Result
valid SOS decrypts correctly	PASS
location is not visible in the radio packet	PASS
same packet sent again is rejected (replay)	PASS
one flipped bit in ciphertext is rejected	PASS
modified header (seq) is rejected	PASS
packet made with wrong key is rejected	PASS
unregistered device ID is rejected	PASS
ACK + message reaches victim and matches last SOS	PASS
packets after victim reboot are still accepted	PASS
truncated packet is rejected	PASS

4. Packet size and airtime

37 bytes SOS packet (encrypted)	62 bytes ACK + 25-char message	≈0.34 s / 2.5 s SOS airtime at SF9 / SF12
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Airtime was calculated with the Semtech LoRa time-on-air formula (BW 125 kHz, CR 4/7, 8-symbol preamble, explicit header, CRC on). The encryption overhead is 23 bytes per packet.

5. Next steps

On-device validation after the hardware arrives: encryption timing on the ESP32, and field tests measuring range, delivery rate, confirmation time and battery life.